BRAC UNIVERSITY
Dept. of Computer Science and Engineering

Quiz 1

CSE428 : Image Processing
Total Marks: 24
Time: 30 minutes

Student ID:		**Section:**	

1. Answer the following questions : **[8]**

   (a) What is histogram equalization? List TWO advantages and ONE limitation of this technique? **[3]**
   (b) Define Weber's ratio in image processing. Write its mathematical formula and explain its significance in human perception. **[3]**
   (c) Draw rough sketches of transformation functions for: (a) Negative transformation (b) Log transformation **[2]**

2. A surveillance camera captured an image with the following 4×4 grayscale pixel matrix (3-bit intensities): **[16]**

1	2	3	1
0	1	5	4
2	1	5	2
1	0	2	3

A. Create a comprehensive table showing the intensity levels (r), frequency count, PDF, and CDF for each intensity level present in the image. Also, draw the histogram based on your frequency data.
[8 + 2]

B. Based on the histogram, comment on the overall brightness and contrast of the image. Justify why histogram equalization is suitable for this image.
[2]

C. Create the transformation function for histogram equalization. List the transformation values for each intensity level and draw the transformation function graph plotting T(r) vs r (output intensity vs input intensity). Mark all transformation points clearly on the curve. **[4]**

D. Apply the transformation function to each pixel in the original image matrix to generate the equalized intensity values and construct the enhanced image matrix.